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Next-generation specimen digitization: capturing reflectance spectra from the world's herbaria for modeling plant biology across time, space, and taxa

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Summary

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Spectral reflectance measured from herbarium specimens represents a potentially vast source of information relevant to plant taxon identification and functional traits, which has inspired many laboratories world-wide to initiate next-generation spectral digitization from specimens. Combining these datasets into a coordinated global database would generate new capacity to model plant traits globally, enabling connection with remote sensing and ecological and biosphere models, as well as reconstruction of trait evolution. However, coordination is needed to avoid downstream problems in data aggregation due to variation in data standards and technical specifications of the instruments, optical setups, or measurement protocols. The International Herbarium Spectral Digitization (IHerbSpec) working group has initiated a globally collaborative program, outlining the central issues to address in establishing protocols, standards, and best practices, and proposing next steps. This collaborative effort will allow generation of replicable spectral reflectance data from plant specimens housed in herbaria around the world within ongoing digitization programs following community-defined standards and Findable, Accessible, Interoperable, and Reusable (FAIR) principles.

I. Introduction

In an era of rapid global change and biodiversity loss, safeguarding our knowledge of plant diversity is essential. Herbaria serve as foundational repositories for this knowledge, both through their traditional applications to document species morphology and distribution as well as novel applications of digitization and technology (National Academies of Sciences, Engineering and Medicine, 2020; Heberling *et al.*, 2021; Davis, 2023; Mandrioli, 2023). Advancing technologies have now enabled investigations that apply whole-genome sequencing and metabolomic methods to herbarium specimens that are up to hundreds of years old (Burbano & Gutaker, 2023; Davis, 2023; Medeiros *et al.*, 2024; Davis & Knapp, 2025). These advances are now routinely used in macroecological and biogeographical studies, providing a vast botanical record of species distributions in space and time relevant to monitoring biodiversity change in the Anthropocene (Willis *et al.*, 2008, 2017; Meineke *et al.*, 2018a,b; Pearson *et al.*, 2020). Here, we broaden the extended specimen concept, which re-envision the role of specimen data as a vast, connected repository of information about individual organisms (Webster, 2017; National Academies of Sciences, Engineering and

Medicine, 2020), to include spectral reflectance from dried plant specimens. The contribution of spectral information to obtaining both functional and phylogenetic information in plants centers on reflectance spectra as a critical data type in the plant sciences with high potential to integrate information about plant diversity across scales, from leaves to ecosystems and the biosphere (Jetz *et al.*, 2016; Cavender-Bares *et al.*, 2017, 2025; Wang *et al.*, 2022; National Academies of Sciences, Engineering and Medicine, 2025). When combined with even a fraction of the world's 400 million herbarium specimens (Thiers, 2024), reflectance spectroscopy provides a new means to extend our inferences of plant phenotypic and functional variation across space and time – and across the entirety of plant taxonomic and phylogenetic diversity (Kothari *et al.*, 2023; Kühn *et al.*, 2024; Neto-Bradley *et al.*, 2026; White *et al.*, 2025b).

As with digitization standards for herbarium imaging over the past 25 years (Nelson *et al.*, 2015; Hedrick *et al.*, 2019; Davis *et al.*, 2021), similar consensus on protocols and standards is now needed to incorporate spectral data into the 'global metaherbarium' (Davis, 2023). Standardized workflows for spectral data and metadata collection will enable integration across herbarium collections, linking trait estimates to individual specimens and

the time, place, and environment in which they were collected (Davis *et al.*, 2015; Willis *et al.*, 2017; Meineke *et al.*, 2018a,b; Pearson *et al.*, 2020). This article documents the purpose and vision of the newly established international herbarium spectral digitization (IHerbSpec) working group and our progress to date in advancing spectral digitization within herbaria across all green plant groups as a scalable tool for biodiversity science. We describe the nature of spectra and the prospect of measuring them in the world's herbaria, highlight the benefits of this massive effort, and address key challenges and next steps.

II. The expanding use of plant traits in ecology and evolution

Extensive work has been carried out by plant systematists to describe new species based on plant traits and to infer rates, processes, and order of trait evolution, often in conjunction with genetic and genomic data (e.g. Beaulieu *et al.*, 2012; Scoffoni *et al.*, 2016; Delaux *et al.*, 2019). Plant traits have become fundamental to understanding ecological processes such as interactions with the abiotic environment, interactions with herbivores and pollinators, growth responses to resource availability, community assembly processes, and contributions to ecosystem and biosphere functions (Lavorel & Garnier, 2002; Violle *et al.*, 2007; Cavender-Bares *et al.*, 2016a; Funk *et al.*, 2017; Shipley *et al.*, 2017; Dechant *et al.*, 2024). Traits provide key insights into consistent patterns of resource acquisition and ecological functions across the green plant tree of life, termed the Plant Economics Spectrum (Wright *et al.*, 2004; Díaz *et al.*, 2016). These patterns influence species distributions across environmental gradients and inform models of community assembly, vegetation dynamics, and biosphere function (Cavender-Bares *et al.*, 2016a). Consequently, significant effort has been invested in developing global plant trait databases (e.g. TRY (Kattge *et al.*, 2020), BIEN (Enquist *et al.*, 2016), AusTraits (Falster *et al.*, 2021; Wenk *et al.*, 2024)) and scaling plant traits to global maps to model biosphere dynamics (Dechant *et al.*, 2024). Frequently collected functional traits for leaves include nitrogen and other nutrients, leaf mass per area, cellulose, lignin, pigments, stable carbon isotopes, and other biochemical compounds, such as carbohydrates, proteins, lipids, and secondary metabolites (Fig. 1). Plant functional traits are well-studied and measured in temperate and arctic regions in well-resourced, easily accessible regions of North America, Europe, China, and Australia, but are undersampled in tropical regions (Jetz *et al.*, 2016; Bjorkman *et al.*, 2018). Common and widespread species are much better represented in plant functional trait databases than rare species, seasonally ephemeral species, and species from understudied biogeographic regions. Given that herbaria provide access to plant specimens collected by generations of botanists, they offer a means to systematically obtain trait data from plants in taxa that are rare and/or occur in ecosystems or regions of the world that are difficult to access, filling in global gaps in plant functional traits of the known plant species and lineages (Heberling, 2021). Although sampling biases also exist within herbaria (Daru *et al.*, 2018), their coverage is better than that of trait databases.

III. What are reflectance spectra?

Reflectance spectra are measurements of electromagnetic radiation (light) returned from surfaces to a sensor, resolved to hundreds of narrow wavelength bands that are each only a few nanometers in width (Fig. 1) (Box 1). Plant tissues from across the tree of life have varying structural, anatomical, organellar, and biomolecular compositions that influence their spectral properties (Elvidge, 1990; Gitelson & Merzlyak, 1994; Sims & Gamon, 2002; Ustin *et al.*, 2004; Asner & Martin, 2011; Serbin *et al.*, 2014; Cavender-Bares *et al.*, 2017; Jacquemoud & Ustin, 2019; Serbin & Townsend, 2020), and spectral data have been widely used to estimate structural, chemical, and physiological plant traits (Ustin *et al.*, 2009; Serbin *et al.*, 2012; Féret *et al.*, 2017; Costa *et al.*, 2018; Prananto *et al.*, 2020; Chlus & Townsend, 2022; Wang *et al.*, 2023; Supporting Information Table S1). Spectra can also provide information about taxonomic identity (Durgante *et al.*, 2013), phylogenetic placement (Meireles *et al.*, 2020b), defense chemistry in diverse genera (Fine *et al.*, 2021), and phenotypic variation linked to genetic or phylogeographic variation (Cavender-Bares *et al.*, 2016b; Deacon *et al.*, 2017; Stasinski *et al.*, 2021; Hernández-Leal *et al.*, 2025), and provide a holistic view of plant phenotypes (Kothari & Schweiger, 2022). Reflectance spectra spanning visible (VIS), near-infrared (NIR), and short-wave infrared light (SWIR) (400–2500 nm) from freshly harvested leaves differ from those obtained from pressed and dried leaves because of changes in water content, pigments, and structure during drying (Fig. 1). Spectra from fresh leaves can provide accurate estimates of plant pigments, traits related to water content, and mobile, degradable, or small-molecular-weight carbon compounds (Wang *et al.*, 2023). Dry leaves often provide more accurate predictions of macronutrients, micronutrients, and carbon or nitrogen stable isotopes (Chlus & Townsend, 2022; Kothari *et al.*, 2023) but are unsuitable for predicting water-related traits and some pigments that degrade during drying (Ustin *et al.*, 2009).

The VIS range (400–700 nm) is strongly associated with Chl, carotenoids, anthocyanins, and other pigments (Lichtenhaler, 1987; Gitelson *et al.*, 1998; Gamon & Surfus, 1999). Chl is partially degraded when exposed to light and may be reduced in herborized leaves, but it can still be detected. Spectral information related to carotenoids, flavonoids, and other chemical compounds remains readily detected and may be less susceptible to degradation. In the NIR (1000–2500 nm), vibration frequencies associated with molecular functional groups (–CH, –NH, –OH) show features of primary metabolites (carbohydrates, lipids, and proteins) as well as secondary metabolites (phenolic compounds, terpenoids, and alkaloids) (Türker-Kaya & Huck, 2017; Jacquemoud & Ustin, 2019; Ustin & Jacquemoud, 2020). Spectral regions beyond 2500 nm reveal additional information. In the mid- and long-wave infrared regions (2500–25 000 nm), for example, detailed chemical identification is possible, particularly for pure chemical compounds (Türker-Kaya & Huck, 2017), providing opportunities to detect ecologically and evolutionarily important differences among organisms. Measurement of reflectance across these spectral regions can inform taxonomic determinations (Sousa da Silva & Hopkins, 2024).

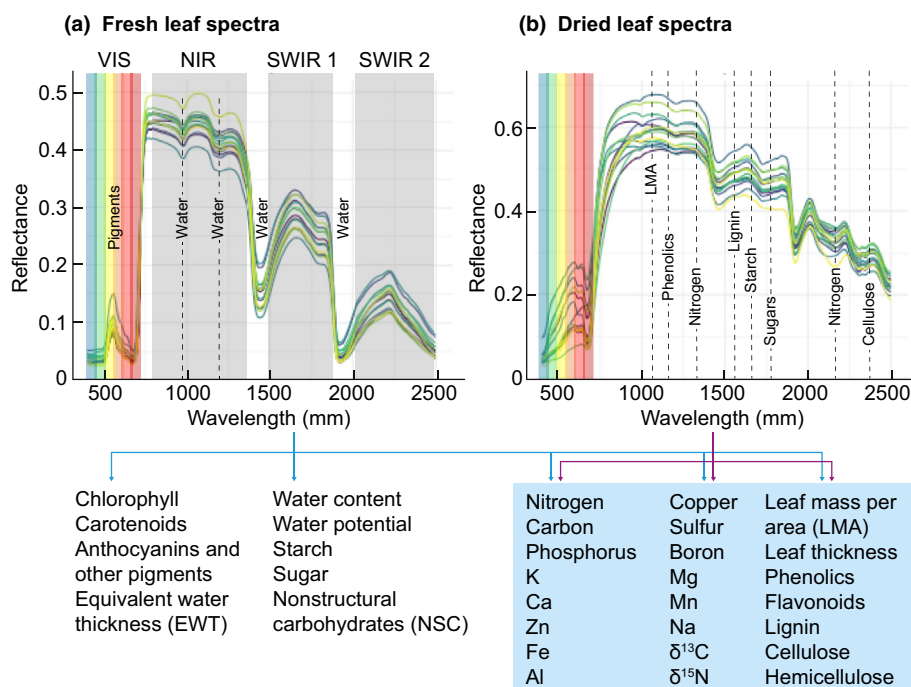


Fig. 1 Plant traits derived from full-range spectral reflectance (400–2500 nm) from fresh leaf tissue (a) and dried, pressed leaves (b). Spectral regions can be characterized into the visible range (VIS, 400–700 nm, shown in rainbow colors in both a and b), the near-infrared (NIR, c. 700–1100), and the short-wave infrared (SWIR1, c. 1100–2000, and SWIR2, c. 2000–2500), shown by the gray-shaded areas in (a). Plant pigments (chlorophyll a and b, carotenoids, anthocyanins), traits related to water content and water potential, including leaf dry mass concentration, equivalent water thickness, and mobile, degradable or small-molecular-weight carbon compounds (sugars, nonstructural carbohydrates) can be derived from spectra in fresh leaves. In dry leaves, water-related and pigmentation traits may not be reliably derived due to desiccation. In both fresh and dry leaves, it may be possible to estimate macronutrients (nitrogen (N), carbon (C), phosphorus (P), and potassium (K)) and micronutrients (calcium (Ca), zinc (Zn), iron (Fe), copper (Cu), sulfur (S), boron (B), magnesium (Mg), manganese (Mn), aluminum (Al), and sodium (Na)), as well as large stable carbon-based molecules (phenolics, lignin, cellulose, or hemicellulose), defense compounds (phenolics, flavonoids), and stable isotopes ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$). Indeed, they may be more accurately detected from dried tissue due to the absence of water. Water absorbs energy throughout the spectral range shown, but particularly in the 700–2500 nm bands indicated in (a). Some of the spectral features that are important for predicting traits are indicated in (b), adapted from Cavender-Bares *et al.* (2025).

Given their utility in detecting variation in plants, reflectance spectra are widely used in applied contexts, including remote or proximal detection of heavy metals and other environmental contaminants in plant tissues (Zhang *et al.*, 2022), pathogen infections (e.g. Fallon *et al.*, 2020; Sapes *et al.*, 2024), or the spread of invasive species (e.g. Hestir *et al.*, 2008; Khanna *et al.*, 2012; Santos *et al.*, 2016; Dao *et al.*, 2021; Mallmann *et al.*, 2023). High-throughput methods of spectral phenotyping are increasingly used in plant breeding programs (e.g. Ge *et al.*, 2016; Meacham-Hensold *et al.*, 2019). Major efforts are underway to scale up spectral data from plants for biodiversity monitoring (Jetz *et al.*, 2016; Cavender-Bares *et al.*, 2020, 2022; Wang *et al.*, 2020; Williams *et al.*, 2021; Blanchard *et al.*, 2024; Dechant *et al.*, 2024; Hadlich *et al.*, 2025).

IV. Estimating plant traits from spectra of herbarium specimens

Advancing technologies and a suite of studies over the past decade have set the stage for applying spectral biology to herborized specimens to predict plant function (Fig. 1). Some of the first spectral studies of dried leaves were conducted in the early 1990s and contributed to early vegetation index-based or radiative

transfer model-based approaches to linking spectra with traits (Elvidge, 1990; Fourty *et al.*, 1996). More recently, Costa *et al.* (2018) showed the potential of using machine learning models trained on spectra on silica-dried leaves to predict traits of tropical tree species. Building on this work, Kothari *et al.* (2023) showed similar results for a wide range of traits in a large data set of unmounted pressed leaves from temperate forest species. Sampling these same species, White *et al.* (2025b) subsequently showed that spectral models trained using detached leaves from decades-old herbarium specimens accurately predicted key traits such as leaf mass per area. Kühn *et al.* (2024) applied a workflow to predict leaf traits from herbarium specimens under different storage conditions to demonstrate the potential to track temporal trends in leaf carbon, nitrogen, and phosphorus content.

V. Applying spectral signatures to phenotype–genotype associations and evolutionary models

Coupling phenotypic data from spectroscopy with genetic and genomic information (Fig. 2) can help determine ecological factors and evolutionary processes involved in shaping underlying traits (Matsuda *et al.*, 2012; Čepel *et al.*, 2018). Spectral data can reveal

Box 1. Brief glossary of terminology applied to plant reflectance

Band: A range of wavelengths within a spectrum for which the radiation is detected and reported as a single variable.

Bandwidth: The wavelength range or span of the spectral band.

Dark current: A small electric current in a spectroradiometer independent of incident light.

Diffuse reflectance: Light reflectance scattered in many directions due to surface roughness that is largely independent of viewing angle. In leaf spectroscopy, this signal carries information about internal structure and chemical composition.

Electromagnetic spectrum: Forms of wave energy described in terms of wavelength or frequency. Shorter wavelengths have higher energy, and longer wavelengths have lower energy.

Fiber optic: Strands of glass or plastic (optical fibers) that transmit light from the end of the strand to the spectrometer detector.

Full-range spectrum: Electromagnetic energy spanning the range of 400–2500 nm, covering the visible (VIS), near-infrared (NIR), and short-wave infrared (SWIR); commonly used in foliar spectroscopy.

Herbarium specimen: A preserved plant or part of a plant, lichen, alga, fungus, or other material with associated taxonomic, geographic, temporal, and other collection metadata stored in an herbarium.

Herborization: The set of processes by which a plant (or other) specimen is collected, prepared, preserved, and incorporated into a permanent herbarium collection facility, together with the physical and chemical changes it undergoes in its preservation and storage environment through time.

Hyperspectral: Narrow bandwidths for all bands across an interval of the electromagnetic spectrum; in leaf spectroscopy, spanning the VIS to NIR or VIS to SWIR spectrum.

Indium gallium arsenide (InGaAs) sensor: Material sensitive to NIR and SWIR electromagnetic energy, typically from 900 to 1700 nm, and up to 2500 nm depending on composition.

NIR: The range of wavelengths used in foliar spectroscopy in the electromagnetic spectrum from c. 700 to 1100 nm, the wavelength region between VIS and SWIR.

Partial Least Squares Discriminant Analysis (PLS-DA): A supervised machine learning approach used for classification based on high-dimensional datasets such as spectral data. It is a variant of partial least squares regression with a categorical response variable.

Partial Least Squares Regression (PLSR): A statistical method that generates latent variables from predictors (X) (e.g. spectral data) designed to capture information relevant for predicting the responses (Y) (e.g. foliar traits). The latent variables are linear combinations of the original predictors that maximize their covariance with Y. The final model regresses Y on the latent variables, making PLSR well-suited for many, highly correlated predictors.

Pressed plant: A dried specimen that was pressed flat to dry.

Radiative transfer models (RTM): Physical models using computer programs to simulate the reflectance, transmittance, and absorption of solar radiation in various media, including leaves. Governed by physical laws, RTMs can operate in forward mode to predict spectral responses of leaves based on material characteristics or in backward mode to infer material properties from leaf spectra. These models vary in complexity, balancing computational demand, accuracy, and scalability. The number of traits that can be simultaneously identified is limited. For remotely sensed imagery, inputs such as solar radiation and elevation angle and parameters such as leaf angle distribution and leaf area index help to solve the radiative transfer equation in optical or thermal domains.

Reflectance standards: A material that produces absorption and scattering reflectance of known values, used to calibrate or test the accuracy of a spectrometer's measurements.

Spectral reflectance: Fraction of incident radiant flux that is returned (reflected) across a specified wavelength region of the electromagnetic spectrum.

SWIR: The range of wavelengths used in foliar spectroscopy in the electromagnetic spectrum, typically between 1500 and 2500 nm.

Spectral biology: The study of biological processes and biodiversity using spectrally resolved observations of light's interaction with biological systems to reveal their chemistry, structure, and function across scales, from genomes to ecosystems.

Spectroradiometer: Instrument to measure intensity of electromagnetic energy in spectral bands across a larger interval of wavelengths, for example VIS–NIR (400–1100 nm) or VIS–SWIR (400–2500 nm). Used interchangeably with spectrometer in this article.

Spectroscopy: The study of interactions between electromagnetic radiation and matter; used for vegetation science to analyze the spectral properties of leaves, canopies, ecosystems, and landscapes over space and time.

Spectral resolution: The wavelengths of light within a spectral band; for spectrometers, it is defined as the Full Width of the band when the intensity of the signal is at Half the Maximum intensity measured (FWHM), specifying the short and long wavelength edges of the band. Higher spectral resolution, characterized by narrower bands or smaller FWHM values, allows for finer discrimination of spectral features and thus more detailed identification of materials.

Spectral range: Span between the smallest and largest wavelengths that a sensor or system can detect.

Specular reflectance: Mirror-like reflection of light from a smooth surface relative to the size of the photon, in which the light reflects from the surface at the same angle and plane to the normal as the incident ray, but in the forward direction.

Taxon discrimination methods: Analytic methods aimed at classifying unidentified samples into taxonomic categories, or clustering samples according to taxonomic affinities.

Trait models: Physical, statistical, or machine learning models used to predict foliar traits from leaf spectra.

VIS: The range of the electromagnetic spectrum associated with human vision from 400 to 700 nm.

variations in traits relevant to ecological niches, while genomic data provide insight into the genetic architecture underlying these traits (Madritch *et al.*, 2014; Blonder *et al.*, 2020). Together, these kinds

of datasets can reveal how populations adapt to environmental pressures. For instance, closely related species of *Quercus* subsection *Virentes* show spectral phenotypic divergence associated with

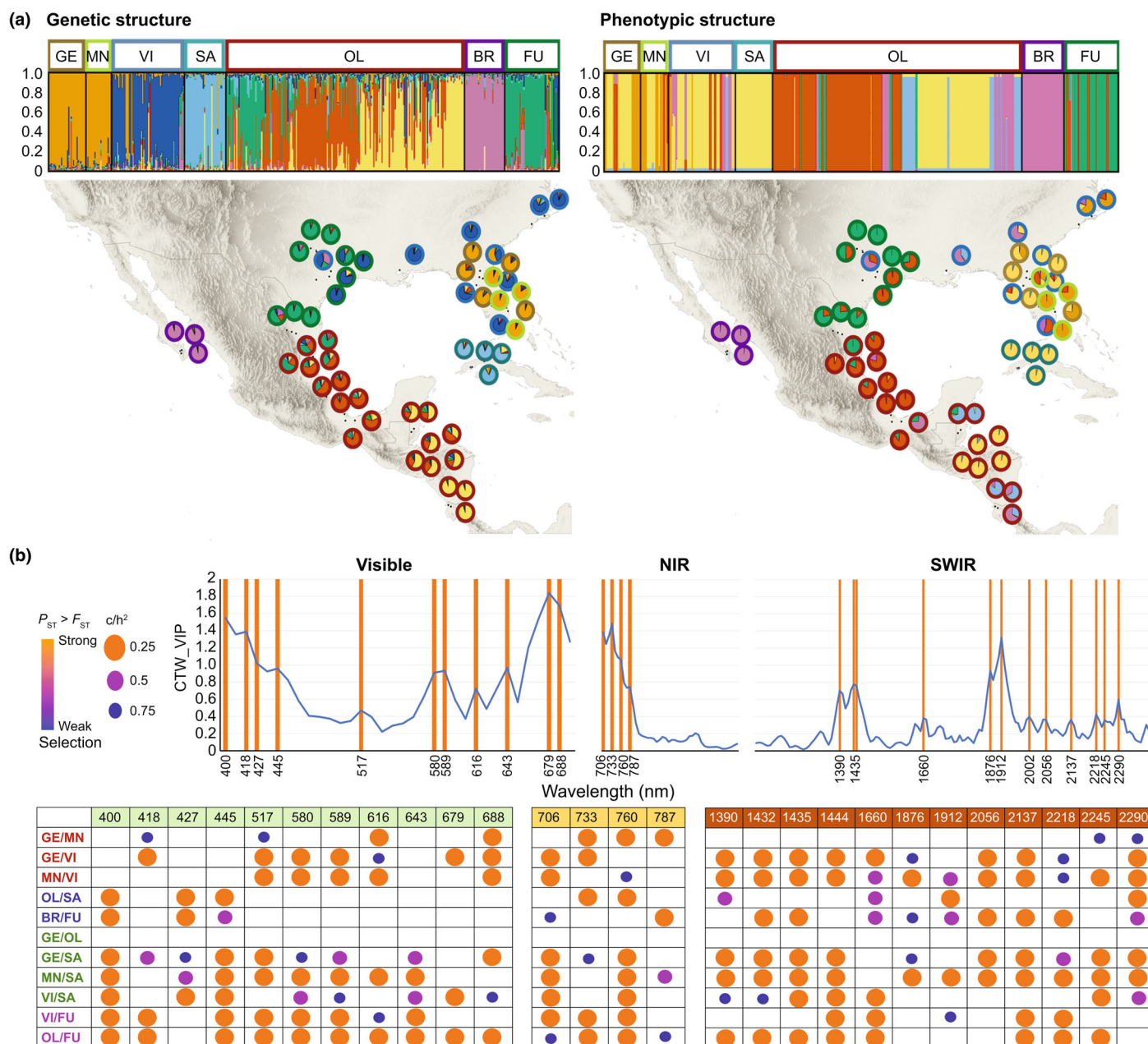


Fig. 2 An example of combining genotypic and phenotypic data derived from pressed leaf spectra to compare spatial genetic and phenotypic structure as well as pairwise phenotypic and genetic distances (P_{ST} vs F_{ST}) to test for selection. (a) The spatial distribution of genetic (left) and phenotypic (right) variation across populations of the seven live oak species in *Quercus* section *Virentes*. Seven genetic groups were identified using STRUCTURE (Pritchard *et al.*, 2000) using nSSRs from 56 populations, and six phenotypic groups were identified by GENELAND (Guillot & Santos, 2009) using spectrally derived leaf traits (leaf mass area, thickness, solubles, hemicellulose, cellulose, and lignin) from dried specimens from 423 individuals. The percentage assignment to genetic or phenotypic groups is shown at both the individual tree level (upper bar plots) and subpopulation level (pie charts). Colors inside the pie charts represent genetic or phenotypic groups: BR, *Quercus brandegeei*; FU, *Quercus fusiformis*; GE, *Quercus geminata*; MN, *Quercus minima*; OL, *Quercus oleoides*; SA, *Quercus sagraeana*; VI, *Quercus virginiana*. (b) P_{ST} and F_{ST} comparisons between pairs of sympatric or sister species using selected spectral bands (orange vertical lines) within the VIS, NIR, and SWIR regions. Spectral bands were selected based on their importance in discriminating among species using PLS-DA (Variable importance of projection, VIP, black line). Matrices show the strength of selection for different wavelengths and pairwise species comparisons based on the magnitude of difference between P_{ST} and F_{ST} values. P_{ST} and F_{ST} comparisons are plotted as a function of the additive genetic variation (c) to estimated heritability (h^2) ratio, for the optimal value of c/h^2 at which the lower confidence limit of P_{ST} is higher than the upper confidence limit of F_{ST} depicted at 0.25 (strong selection, orange) to 0.75 (weak selection, blue). Genetic data are from Cavender-Bares *et al.* (2015). Figure is adapted from Hernández-Leal *et al.* (2025).

specific microhabitats, shaped by local adaptation to ecological variation (Fig. 2; Hernández-Leal *et al.*, 2025). Comparisons of spectral data with genomic markers have been used to determine genetic loci associated with key traits, such as cold tolerance or drought resistance, as well as how they vary across populations (Madritch *et al.*, 2014; Cavender-Bares *et al.*, 2016b; Czyż *et al.*, 2020). Spectral phenotype–genotype associations offer a robust framework for uncovering the interplay between natural selection, demographic processes, and adaptation (Kokaly *et al.*, 2009; Cavender-Bares *et al.*, 2016b; Deacon *et al.*, 2017; Blonder *et al.*, 2020; Stasinski *et al.*, 2021; Hernández-Leal *et al.*, 2025) and for deciphering evolutionary processes across the plant tree of life (Meireles *et al.*, 2020a).

Spectral data can also be used to model species traits on phylogenetic trees using comparative methods (Harvey & Purvis, 1991; Cavender-Bares, 2019; Meireles *et al.*, 2020a). When spectra are consistently measured across species, spectrally derived traits (or spectral indices and variables themselves) can be used to model evolutionary rates, constraints to evolution, and selection, by fitting evolutionary models to these species-level variables in relation to a phylogenetic tree (or trees) using phylogenetic comparative methods. These approaches are widely used in comparative biology, including for the study of plant trait evolution (e.g. Ackerly, 2009; Beaulieu *et al.*, 2012; Cavender-Bares *et al.*, 2018; Fontes *et al.*, 2025), to discern shifts in trait optima across the phylogeny in relation to geography or environment, to test for convergent evolution, or to test for the correlated evolution of traits and species environmental niches (Edwards, 2006). Applying these approaches to spectra and spectrally derived traits is an area of active research. In a simulation exercise, we show how dried leaf spectra are expected to evolve according to four models of evolution: Brownian motion random walk model with fast and slow rates of evolution, a single-optimum Ornstein–Uhlenbeck model of adaptive evolution, and an Early Burst model with declining rates of evolution through time (Fig. 3).

A major challenge, however, is the high dimensionality of full-range spectral data, which comprise hundreds of interdependent variables per measurement. Incorporating this high dimensionality within a phylogenetic comparative framework is challenging both statistically and computationally because of the high number of coevolving traits in a lower number of taxa (Clavel *et al.*, 2019). Novel comparative methods such as phylomorphospace and penalized likelihood approaches will likely have useful applications to spectral data in phylogenetic frameworks (Clavel *et al.*, 2015, 2019; Adams & Collyer, 2019).

VI. Taxon discovery and discrimination

Spectral data have become a valuable tool for evaluating and refining taxonomic hypotheses, including in species delimitation, taxonomic revision, and the detection of cryptic diversity. Across taxonomic ranks, spectra from dry leaves have been used to supplement DNA-based methods and support systematic studies, particularly in morphologically complex clades. In the Amazon, researchers have analyzed spectral absorption profiles with a Fourier-transform NIR (FT-NIR) spectrometer on unmounted

leaf samples to distinguish closely related species within the genus *Eschweilera* (Lecythidaceae) (Durgante *et al.*, 2013) as well as among different developmental stages of species in the Burseraceae (Lang *et al.*, 2015). Paiva *et al.* (2021) applied similar methods to pressed fern fronds to classify species in the genus *Microgramma* with over 90% accuracy. Both Prata *et al.* (2018) and Damasco *et al.* (2019) integrated DNA and spectral data to suggest taxonomic solutions in species complexes, including the reestablishment of *Protium cordatum* (Burseraceae) to species rank (Damasco *et al.*, 2019). Spectroscopy is now being used to support new species hypotheses that are primarily based on morphological and morphometric characteristics (Vasconcelos *et al.*, 2020, 2021, 2025; Gaem *et al.*, 2022; Costa *et al.*, 2025) or to differentiate hybrids from parental species (Deacon *et al.*, 2017; Stasinski *et al.*, 2021).

Applications of these approaches to herbarium specimens are more recent. White *et al.* (2025b) showed that spectral models from herbarium specimen leaves of diverse ages and sources (with a median age of 91 yr) could classify specimens to species with up to 74% accuracy and genera up to 84% accuracy. In a broad study of stone oak (*Lithocarpus*) specimens from across Asia, Neto-Bradley *et al.* (2026) found that machine learning models trained on spectra work nearly as well as those from digitized images for identifying taxa. They concluded that spectra may be particularly important for identifying incomplete specimens of historical significance that may otherwise only be identified to family. In the future, it should be possible to combine digital images with spectral profiles to increase our power of discrimination at different taxonomic levels.

For plant systematists, these advances represent a major opportunity to re-explore herbarium collections with new eyes. Spectral profiles can uncover consistent differences in chemical and structural traits in specimens that may not be evident through morphology. Spectral classification models using herbarium specimens can complement morphology and genetics and serve as a test of taxonomic validity, revealing whether phenotypic variation aligns with expected groupings. The success of taxonomic discrimination depends on the similarity of leaf and other plant part morphologies within each of the assigned taxa and can be interpreted as a test of cohesion within groups and distinction between them. High within-taxon variation or convergent phenotypes can reduce accuracy. Moreover, informative spectral regions may shed light on the evolutionary processes shaping trait divergence. By revealing the spectral regions that can best distinguish taxa, discriminatory models may contribute to deciphering how taxa have diverged or converged and the evolutionary forces that have shaped their phenotypes.

Generating spectral measurements from specimens of known identity will enable the development of reference datasets for identification purposes and may help to provide a preliminary taxonomic classification of the multitudes of specimens in herbaria that have not been identified because they lack diagnostic structures or have not yet been studied by sufficiently knowledgeable experts (Neto-Bradley *et al.*, 2026). Fresh leaf spectra show promise for classifying species to all taxonomic ranks (Meireles *et al.*, 2020b; Mallmann *et al.*, 2023; Blanchard *et al.*, 2024; Hadlich *et al.*, 2025), and dried leaf spectra may show even greater promise because the

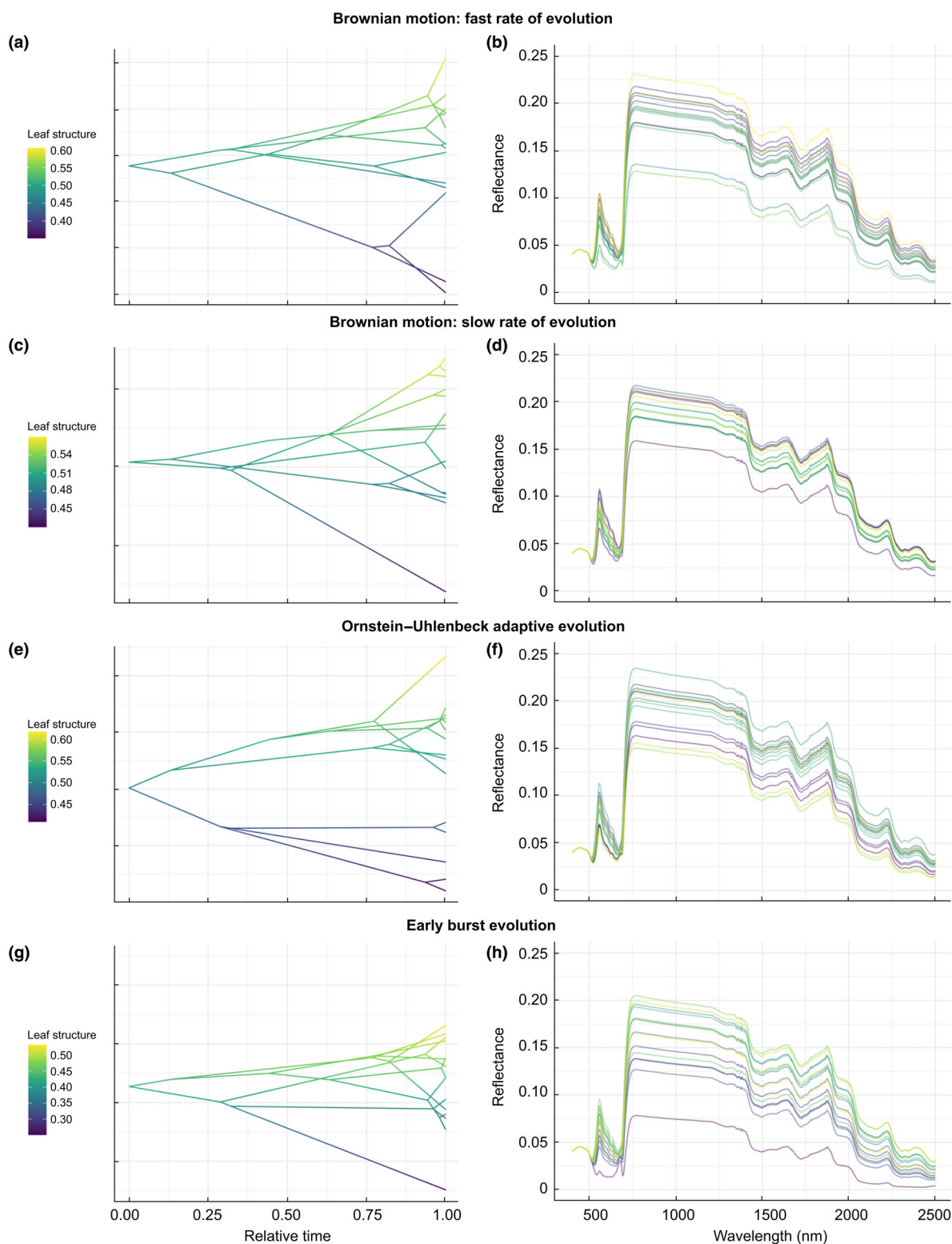


Fig. 3 Simulated evolution of leaf structure (i.e. numbers of cell layers, N) under three models of evolution and the corresponding dried leaf spectra predicted from leaf traits using the PROSPECT D radiative transfer models (Féret *et al.*, 2017) following Meireles *et al.* (2020a). Each row represents one model: Brownian motion with fast (a, b) or slow (c, d) evolutionary rates, a single-optimum Ornstein–Uhlenbeck adaptive evolution process (e, f), and an Early Burst process with declining rates over time (g, h). The graphs on the left show phylogenetic trees through relative time (x-axis) with y-values and branch colors indicating simulated leaf structure values. The graphs on the right show simulated reflectance spectra (400–2500 nm) for the final trait values of each lineage. (Starting values for the traits in the models are as follows: leaf structure, $n = 0.5$; Chl, $Cab = 10$; carotenoids, $Car = 12.0$; LMA/leaf dry matter content, $Cm = 0.0005$; water, $Cw = 0$; sigma values for $n = 0.1$ for all models except for the slow Brownian Motion model, for which it is 0.5).

removal of the water absorption bands in the SWIR may reveal more distinctive absorption features (Kothari *et al.*, 2023). Stepwise hierarchical approaches for classifying taxa first to broader and then narrower clades are a promising approach (Fig. 4), and would sidestep the problem of computational limits and reduced accuracy in discrimination among large numbers of classes. Achieving high accuracy would require highly populated spectral libraries at each phylogenetic scale. We note that some taxa have complexities, such as symbioses in lichens between algal and fungal partners, that will require the development of novel approaches.

VII. Benefits and challenges to aggregating and scaling spectral data

While there is considerable excitement for the collection and application of herbarium spectral datasets to these botanical applications, methodological, material, and data processing challenges remain. While spectral measurements record information about the target phenotype, differences in instrumentation, specimen preservation, and data processing pipelines introduce additional variability that complicates batch processing for large-scale data aggregation and cross-institutional compatibility (Fig. 5; White *et al.*, 2025b). Spectra-based models for ecological and evolutionary inference are typically robust to known and well-represented sources of variation but may produce biased predictions when trained on incomplete or unbalanced datasets that fail to account for the full diversity of biological and technical conditions.

1. Variation in foliar phenotypes

Leaves' tissue properties vary due to a multitude of factors, including leaf age and developmental stage, environmental conditions during growth (e.g. light, water, CO₂, temperature), and a range of other factors (Fajardo & Siefert, 2016; Chlus & Townsend, 2022) across environmental gradients and even within a single tree canopy. All these sources of variation in leaf traits can influence the reflection of light (Stefanski *et al.*, 2025). Within a single leaf, tissue properties will change from the edges to the vascular tissue. With age, leaves add secondary cell wall material, cuticles become thicker and more structurally complex, and concentrations of biochemical compounds change, including compounds that are upregulated or produced in response to stress. Herbarium specimens from species with asynchronous flowering or fruiting may disproportionately contain young, developing leaves or reproductive structures, leading to spectra that do not adequately represent mature leaves (Fig. 6). Predictive models show promise in accommodating such biological variation as long as models are trained to 'see' the full range of variation (Lang *et al.*, 2015; Wang *et al.*, 2023).

2. Adaxial and abaxial leaf surfaces

Adaxial and abaxial surfaces are often morphologically distinct. While many previous protocols oriented toward remote sensing projects only measured the adaxial surface of the leaf because it is the

side observed from above, measuring both leaf surfaces provides a fuller characterization of leaf tissue properties. Ideally, projects will measure spectra from the abaxial and adaxial surfaces of single leaves in tissue packets, plus other leaves attached to the sheet where only a single surface type might be available. Given the differences, it is important to document which leaf surface was measured in each reading (Ustin *et al.*, 2001).

3. Leaf size

Leaves smaller than the measurement area of the optical setup cause reductions in the signal-to-noise ratio (SNR), and leaves with nonflat surfaces introduce variation in the angle of the light source relative to the leaf surface. Conifer needles, vascular plants with tiny leaves, and other groups such as mosses and lichens can be difficult to measure due to their small size and complex surfaces (Fig. 6). This can be addressed with adaptations to optical detection arrangements, such as fine leaf probes, small radius fiber optic cables, and/or small detection windows, but these modifications can also reduce the SNR and quality of the reflectance signal. Where possible, it may help to measure the flattest parts of the leaves.

4. Specimen age

Specimens undergo chemical and structural degradation through time, even in herbarium cabinets (Burbano & Gutaker, 2023). Although some aspects of this process have been documented, including the breakdown of Chl and accumulation of brown degraded pigments (Fourty *et al.*, 1996), in general, little is known about the suite of changes through time and the factors that slow or accelerate them. White *et al.* (2025b) found that older specimens exhibit a slight decline in correct classification probability from spectra. At our current stage, recording age and other specimen metadata will permit the development of strategies to account for specimen age as a proxy for degradation when integrating spectral datasets (Durgante *et al.*, 2013; Lang *et al.*, 2015; White *et al.*, 2025a).

5. Variation in preservation among specimens in herbarium collections

Herbarium collections capture taxonomic and geographic diversity, often representing centuries of collecting across biomes. Yet, specimens vary widely in how they were collected, treated upon intake, and stored over time, with additional impacts from drying methods or chemical additives such as glues, pest-control agents, and other conservation treatments. These herborization processes have consequences for spectral phenotypes (Kühn *et al.*, 2024; White *et al.*, 2025b). Alcohol preservation, widely used in the tropics to prevent fungal growth and rotting in specimens before herborization (Mori, 2011), can dissolve pigments and impact cellular structure (Wellburn, 1994; Bressan *et al.*, 2014). Heterogeneity in preservation, compounded by differences in institutional storage conditions, poses challenges for spectral research. Developing shared protocols and metadata standards (White *et al.*, 2025a) will be essential to ensure reproducibility and

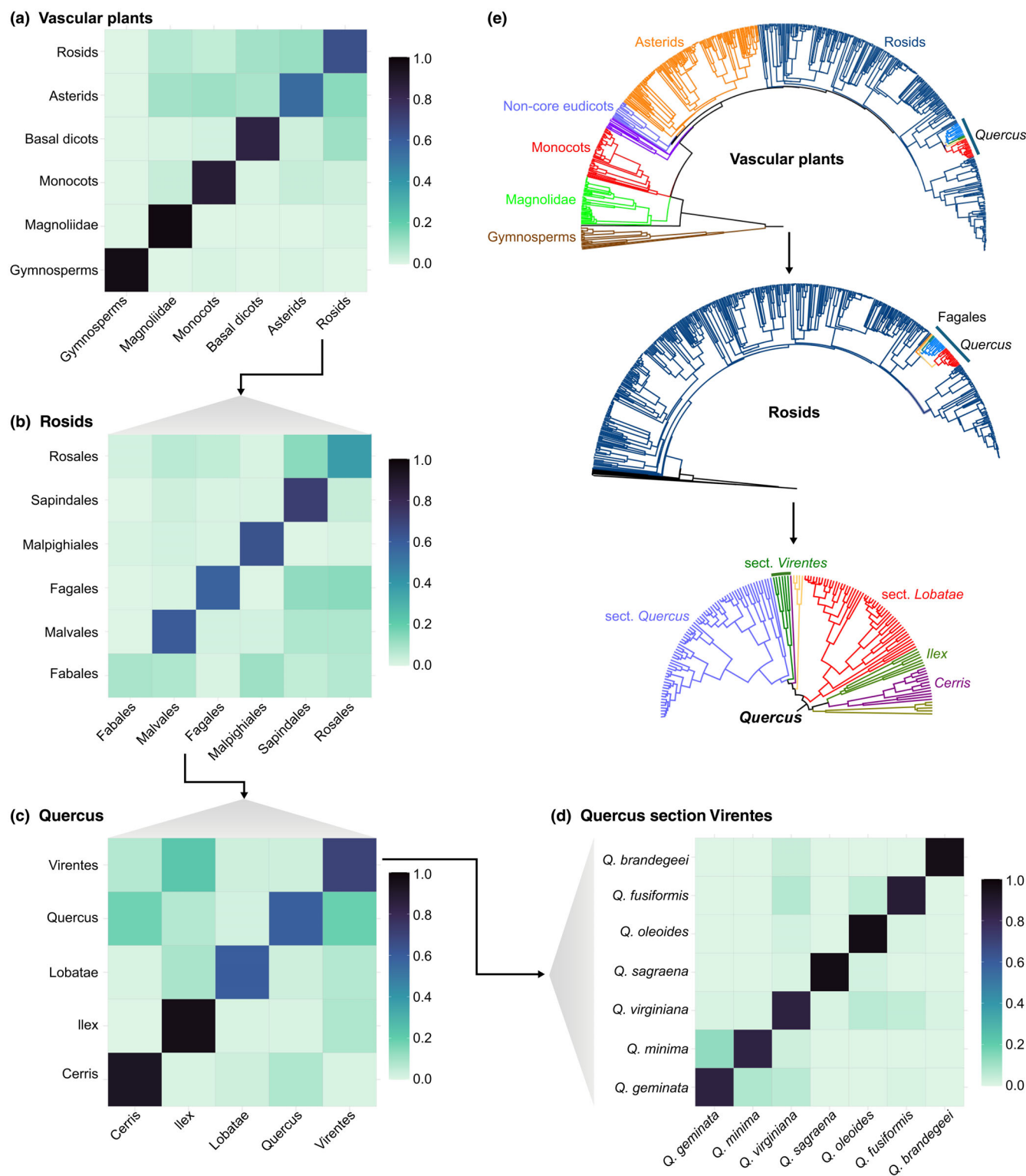


Fig. 4 Stepwise hierarchical approach to taxon classification to place specimens within the plant tree of life. Classification algorithms, such as PLS-DA, are limited by statistical power in the number of entities they can accurately discriminate between. One possible solution is a nested approach where spectral signatures are used to differentiate broad clades within the vascular plants, and stepwise within increasingly smaller clades, such as orders, families, genera – or increasingly narrow phylogenetic lineages. Shown are confusion matrices from broad to increasingly narrow taxonomic groups, starting with (a) broad clades in the vascular plants, including the rosid clade of angiosperms, (b) orders within the rosids, including the Fagales, (c) sections of the genus *Quercus* within the Fagales, and (d) species within *Quercus* section *Virentes*. Correct assignments are on the diagonal, incorrect assignments are in the off-diagonal cells. Data used in (a, b) and the vascular plant phylogeny are from Meireles *et al.* (2020a); spectral data used in (c) are from Cavender-Bares *et al.* (2016b); spectral data in (d) are from Hernández-Leal *et al.* (2025). The *Quercus* phylogeny is from Hipp *et al.* (2020).

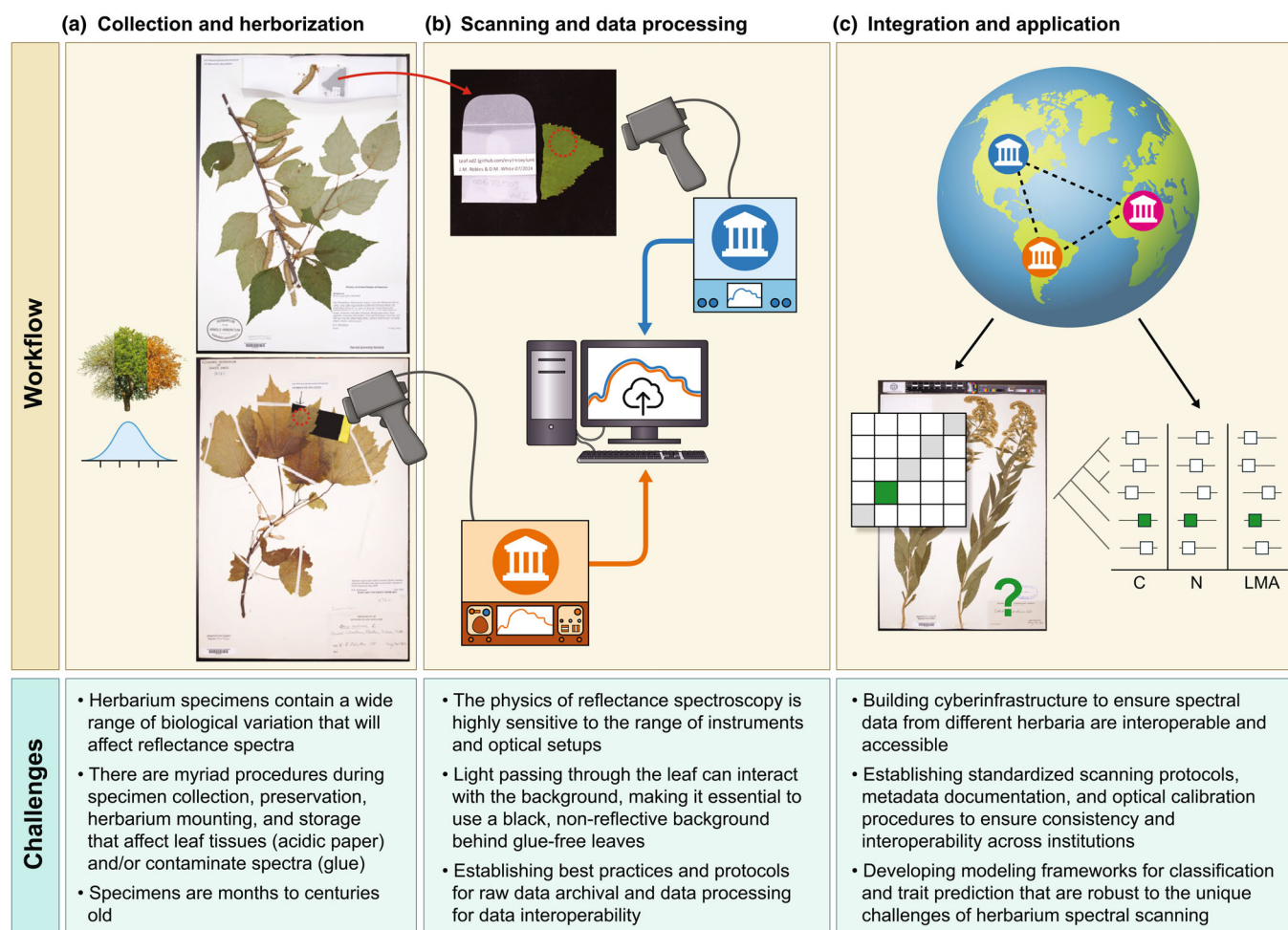


Fig. 5 Herbarium spectral digitization workflow from specimen collection to global integration, and challenges. (a) Variation in specimen condition and mounting methods, including changes in leaf preservation quality, use of adhesives or sewing, and presence of detached leaves in packets. (b) Spectral digitization process using different spectroradiometers, with emphasis on black background placement and data upload; differences in instruments, optics, and data processing. (c) Global integration of spectral data across herbaria, linked to specimen records and trait data such as carbon (C), nitrogen (N), and leaf mass per area (LMA). Photographs are reproduced with permission from the Herbarium of the Arnold Arboretum of Harvard University ((a), upper panel), the Economic Herbarium of Oakes Ames of Harvard University ((a), lower panel), and from the New England Botanical Society Herbarium (c).

cross-study comparability in applications such as trait estimation, taxonomic classification, and evolutionary modeling.

Specimen degradation and quality are strongly influenced by field protocols for pressing and drying. Ideally, plants are immediately pressed in acid-free paper and dried with forced air at ambient temperatures. In practice, however, botanists may rely on ethanol preservatives (70–95%) to prevent rotting under certain conditions and industrial ovens (40–65°C) for drying – methods that can alter tissue integrity and spectral properties.

Additional specimen processing and mounting protocols are applied when dried plant collections enter the herbarium. The use of glue to mount specimens to archival herbarium paper sheets is particularly problematic because the spectral properties of the glue become ‘mixed’ directly with the plant tissue. White *et al.* (2025b) found that its presence reduced the accuracy of species identification from spectral data. While alternatives such as sewing or archival mounting tape avoid this issue, glue remains the most widely used adhesive in the Americas due to its speed and

convenience. Some specimens even contain multiple layers of different adhesives.

Alcohol is frequently used in dehydration. Other herbarium preservation and pest control techniques that can contaminate spectral data include methyl bromide fumigation (now banned) or the application of diatom powder, arsenic, or naphthalene. Computational efforts to ‘unmix’ or ‘subtract’ the contaminant signals from the target tissue spectra have not yet yielded solutions to recovering a clean plant signal (A. Guzmán, B. Neto-Bradley, JE Meireles, unpublished data).

Finally, variation in temperature and humidity through time in the collections environment is another source of variation affecting specimen degradation and spectral properties.

6. Instruments and optics

Variation in detectors and optical setups among instruments creates differences in spectral signatures, adding complexity to the use of



Fig. 6 Variation in the challenges of measuring reflectance spectra on mounted specimens. (a) An example of a relatively straightforward measurement of a specimen leaf against a black background using a one-sided leaf clip with a fiber optic probe with a smaller diameter than the leaf laminar surface area. Isotypes of (b) *Restio arcuatus* and (c) *Vulpia microstachys* are taxa with small leaves. (d) *Abies balsamea*, a conifer with typical narrow needles that is taped to the herbarium sheet but has also been glued, evidenced by the discolored glue at the top of the specimen. A fully taped *Salix arctica* (e) specimen from the Botanical Museum of Copenhagen with conspicuously gray discoloration representing variability in preservation techniques and quality. An herb (f) with delicate, thin leaves that are fully glued to the sheet with no packet containing extra material to measure. A *Sticta* lichen (g) with a nonflat thallus that will distort optical geometry. Circles on the top left demonstrate various aperture sizes (mm) typical of different optical instruments. For comparison, scale bars are shown in (b, c, and f), with a 2-cm bar in (b). Photographs are reproduced with permission from the Herbarium of the Arnold Arboretum (a) and the Gray Herbarium (b, c, and f) of Harvard University, the New England Botanical Society Herbarium (d), the National Herbarium of Denmark (e), and the Farlow Herbarium of Harvard University (g).

herbaria for spectral data collection (Fig. 5; Table 1). The SNR, field-of-view (FOV), and spectral resolution and range are inherent instrument variables that can influence measurements. Standardization of spectral digitization protocols and proper documentation are thus crucial for ensuring interoperability among spectral digitization efforts and integration into broader analytical frameworks (White *et al.*, 2025a). The SNR, for instance, determines the quality and reliability of the measurements, in which higher values mean clearer and more precise spectral data. The pattern of noise varies among instruments and thus influences how well models transfer across instruments. SNR is influenced by light intensity and duration of measurement, as well as reflectance angles (Swayze *et al.*, 2003; Milton *et al.*, 2009). The detector's measurement geometry, instrument calibrations, and wavelength-specific

detector sensitivities particularly affect the high- and low-energy wavelengths in which SNRs may be low (Allam *et al.*, 1987; Swayze *et al.*, 2003). However, the SNR can be increased by increasing the integration time of the spectral measurement and the number of replicate spectral samples measured (Milton *et al.*, 2009). During data analysis, it may be necessary to mask or remove bands with low SNR, for example those at the edges of the spectral range, to improve model performance. However, even noisy bands are found to be increasingly useful with smoothing filters and denoising techniques (Swayze *et al.*, 2003; Aiazzi *et al.*, 2011), including modern machine learning/AI methods (Kim *et al.*, 2021).

The FOV – the angular extent of the observable area that can be seen through foreoptics or lenses – is another variable that often differs between instruments and influences how well specimens are

Table 1 Challenges to aggregating spectral data across specimens and institutions and potential solutions.

Challenge	Description	Mitigation strategy
Variation in leaf phenotypes due to environment	Variation in growth environment (light, temperature, CO ₂ , nutrients) can influence spectra.	Collect spectra on specimens sourced from a range of environmental conditions to capture the variation; avoid midrib and leaf edges, avoid leaves with herbivory and pathogens.
Variation in leaf phenotypes due to leaf age/development stage	Spectral properties and traits change over the lifecycle of the leaf.	Capture information about leaf age in the metadata.
Specimen age	Chemical degradation, pigmentation loss, and exposure to environmental factors over time can influence spectral signatures.	Collect spectra on specimens across a range of ages to capture the full variation. When possible, conduct analyses on specimen age effects. Effects may not be large. Metadata should include date that specimen was acquired and location.
Preservation and storage variation	Drying, decontamination methods, adhesives, and storage environments can influence spectra.	Standardize methods for incoming specimens. Conduct analyses on effects of preservation and storage to determine whether effects are large or small. Increase sample size to capture range of preservation methods.
Alcohol preservation of leaves	Alcohol is used to prevent fungal growth and rotting in specimens before herborization and can dissolve pigments and impact cellular structure	If alcohol was used for preservation, record in the metadata, if known.
Adhesives	Glues contaminate the reflectance spectrum.	Prioritize leaves (or other tissues) without adhesives.
Leaf size	Small leaves are difficult to measure with standard setups.	Fine leaf probes, narrow fiber-optic cables, small detection windows avoid many measurement problems but generally require more measurements to fully sample the leaf spectral variation associated with the species.
Leaf surface	Adaxial and abaxial surfaces differ.	Measure both sides of the leaf. Capture metadata on which side(s) was/were measured.
Foliar surface contamination	Minute epiphytes, bacteria or fungi may be present	Report and describe possible surface contaminants in the metadata
Measuring plant surfaces beyond leaves	Flowers, stems, and other organs are also important.	Measure on a project-by-project basis or using criteria at individual herbaria.
Specimen geometry	Nonflat foliar surfaces generate specular influences on reflectance spectra ('noisy spectra').	Prioritize flat tissues. Increase sample size. Use probe with a small optical aperture.
Instruments and optics	Different instruments give different spectral reflectance signatures. Optical setups also influence spectra.	Adhere to agreed-upon protocols, instrument setup, and metadata standards. Establish interconversion methods. Use the same reflectance standards across all laboratories.
Reflectance background	White mounting paper introduces artifacts into the reflectance spectrum.	Use a black nonreflective background on detached leaves or insert under a mounted leaf that is loose.
Measurement settings and protocols	Sample count, leaf surface choice, integration times, and other user-defined parameters influence SNR, data comparability, and the time it takes to measure a specimen.	Develop minimum sample counts and data standards across herbaria that can be surpassed for specific purposes and/or within individual herbaria.
Destructive sampling for traits	Validation of models for many functional traits (N, P, cellulose, etc.) requires destroying leaf tissue to make direct observations.	Pressed leaves or tissues will need to be sampled ethically and in a coordinated manner.
Metadata collection and access	Data incompatibility due to the range of variables affecting herbarium spectral measurements	Establish metadata standards, similar to Darwin Core format, agreed upon, recorded, and shared by all institutions.

captured in a single measurement. A wider FOV measures a larger portion of the specimen, thereby increasing the representation of within-specimen variability. However, wider FOVs are not ideal for small leaves and make it difficult to target specific leaf regions such as the blade rather than the central vein. Additionally, some instruments present a nonuniform FOV due to the optical fiber bundle and its integration with the light detector, which does not uniformly cover the viewing area (Lévesque *et al.*, 2014). Using instruments with a narrow or nonuniform FOV will require several measurements to adequately capture the variability of optical information.

The spectral range of instruments is distinct from spectral resolution and is commonly used to differentiate between

VIS-NIR (e.g. 350–1000 nm), VIS-SWIR (e.g. 350–2500 nm; also known as 'full range'), MIR (2.5–6.0 μ m), and TIR spectrometers (e.g. 6.0–16 μ m) because these detect photons in different regions of the electromagnetic spectrum. As Fig. 1(b) shows, different traits are inferred from specific regions of the spectrum; thus, measurements that cover the 'full range' (400 nm–2500 nm) are able to capture more of this information. Although VIS–NIR spectrometers are less expensive than full-range spectrometers, some studies on dried leaves show that full-range spectra predict more foliar traits (Kothari *et al.*, 2023; White *et al.*, 2025b). The spectral resolution of many full-range instruments varies within instruments because there are usually two or more light detectors with distinct sensitivity and band

sampling: a silicon sensor (VIS and NIR), an indium gallium arsenide (InGaA) detector for the 1000–1700 nm wavelength region, and an extended InGaAs covering the spectrum to 2500 nm (NIR and SWIR). Due to variation in detector sensitivity and signal availability (i.e. light), silicon detectors are designed for VIS-NIR wavelengths (e.g. 400–1000 nm), in which halogen bulbs generate peak irradiance and tend to have higher spectral resolution than detectors that span NIR-SWIR wavelengths (e.g. 1000–2500 nm). Instruments commonly report data output in 1-nm increments when the true bandwidths are 3–10 nm wide, so their true resolution is 3–10 nm bands. Given the variation in design among instruments, the band center and width for different spectral regions should be recorded in the metadata (will be discussed later).

Additional factors influencing the quality of spectral measurement data include the sensitivity of fiber optics, quality of standards, viewing and illumination angles, and the intensity of light sources (Grant, 1987). Variation in fiber-optic alignment can impact the SNR, requiring careful handling and regular replacement. Manufacturer maintenance should confirm whether all fibers are functional or, if not, determine the percentage of remaining fibers and their locations in the cluster. Use of a standardized leaf probe with integrated illumination avoids problems with the positioning of the lights and the spectrometer. Proper selection of light intensity and integration times is critical to prevent heat-induced changes in specimen integrity and optical properties, and lamps should be operated on direct current rather than on alternating current. Across all of these specifications, regular replacement of calibration standards and routine instrument maintenance, such as cleaning and recalibrating sensors, is essential for maintaining instrument performance.

VIII. Considerations for standardization of protocols and metadata

While detailed protocols and metadata standards remain to be established and agreed upon across institutions, it is important to provide some guidance at this stage, given the rapid increase in the use of spectroscopy for plant studies (Cavender-Bares *et al.*, 2025). Here, we make a series of recommendations and pose questions for consideration. Standardizing spectral digitization setups – including light source positioning, spectral calibration, and reference materials – will be essential to ensure compatibility between datasets collected using different instruments and workflows. IHerbSpec has released a protocol (White *et al.* 2025a, <https://iherbspec.github.io/protocol.html>) for the standardization of measurements and metadata.

1. Sampling

We recommend sampling the variation among leaves on a specimen as well as variation within a single leaf, avoiding leaves with blemishes, contaminants, or pathogens (unless these are relevant to the objectives of the study). When selecting among specimens, prioritizing those where a black, nonreflective background can be readily inserted under some leaves (will be discussed

later) will enable aggregation of ‘clean’ tissue spectral data. Measuring sufficient leaf area to ensure that the spectra are representative of the variation within the specimen should be balanced against the care required to ensure nondestructive sampling. Taking 3–5 measurements per surface across the leaf and 3–5 leaves per specimen can help capture relevant variation, but one fully expanded leaf per specimen may be a sufficient compromise to save time and avoid damage.

2. Reflectance standards

Given the variation in instrument and optical arrangements across institutions, regular measurement of standard cards of known reflectance using a set of certified reflectance standards ranging from black to gray to white provides a means to make data comparable. A rare earth panel can be used to confirm wavelength calibration, especially if the instrument is returned for manufacturer recalibration infrequently. The black and gray standards would be measured regularly, such as once-daily during measurement sessions. The white standard should be measured with each specimen, or at least every two or three specimens if spectral measurements are taken very rapidly.

3. Reflectance background

Light transmitted through leaves may reflect from the background (glue, paper, and even laboratory benches), the degree of this contamination depending on the optical thickness of the leaves and the reflectivity of the background. To minimize this effect, tissues should be measured against a background with < 4% reflectance and low specular reflectance. Suitable materials include black cardstock coated with Krylon® Camouflage Matte Black spray paint and Musou Black USA® IR Flock Sheet, or the manufacturer-supplied black backgrounds for leaf clips (e.g. from ASD, Spectra Vista Corporation, or Spectral Evolution). Background materials should be scanned periodically, digitally connected to tissue spectral data, and archived for reference.

Variation in reflectance background may create challenges for some specimens, which are often affixed to paper. If the specimen has been sewn or taped, it should be possible to slide a thin black sheet between the attached leaf and the paper. Herbarium specimens with loose leaves available in packets may be selected, and those leaves should be checked for glue before being measured against a black background. The importance of using a nonreflective black background will likely prevent automation through conveyor belts, as has been used for the digitization of specimen images.

4. Metadata standards

Robust metadata are essential for building reliable spectral models, as they ensure transparency across the many biological, preservation, and instrumentation variables that affect spectral data. Standardized metadata enable researchers to identify, quantify, and account for sources of variation, improving the performance and generalizability of models for ecological and evolutionary

inference. Metadata standardization is also critical for harmonizing datasets, as it facilitates the integration of spectral data with specimen-level information such as taxonomy, collection locality, phenology, and specimen characteristics (e.g. preservation quality, presence of micro-epiphytes). Metadata describing specimen preservation method, mounting methodology (e.g. glue), and background help researchers assess potential artifacts and make informed decisions when aggregating or filtering datasets. Projects should also capture instrument details (brand, model, spectral resolution, wavelength interval, calibration history) and measurement materials and protocols (e.g. use of white and black reference standards). The IHerbSpec Protocol v1.0 (White *et al.*, 2025a) is a first attempt to establish a standardized yet flexible metadata template for recording tissue, specimen, instrument, and project-specific fields.

Metadata curation and cyberinfrastructure development will be critical for integrating herbarium spectral data into global biodiversity platforms. By adopting common metadata schemas and persistent identifiers (DOIs), researchers can link spectral data directly to digital databases, fostering seamless collaboration and data reuse. Experience gained from successful protocol standardization and data aggregation initiatives (e.g. Darwin Core and iDigBio (Wieczorek *et al.*, 2012; Soltis, 2017)) can be leveraged to implement a strategy for herbarium spectroscopic data. Advances in spectral data repositories, such as linking reflectance data to GBIF, *speciesLink*, and iDigBio records, will enhance accessibility and ensure that spectral datasets are fully interoperable with existing biodiversity databases (Heberling, 2021; Davis, 2023), meeting the goals of the extended specimen concept (Lendemer *et al.*, 2020).

5. Data processing

Various processing steps may be applied to raw spectra to enhance informative spectral features, reduce noise, or reconcile data from different instruments. These include resampling, normalization, or transformation using derivatives or continuous wavelet transforms. Band resampling is often used to produce a consistent set of wavelength bands to use as variables in analyses, especially when aggregating across measurement platforms. Resampling to higher than the true resolution (e.g. 1 nm) is often done automatically by instrument software, but could introduce artificial, interpolated data to the spectrum. Resampling to a coarser resolution (e.g. 5 nm) is frequently done to help reduce band redundancy for predictive models, but could result in loss of small absorption features.

In full-range instruments, the 350- to 400-nm region and 2400- to 2500-nm regions are often trimmed due to the low SNR in those regions. Reflectance values at the wavelengths where the silicon and InGaA sensors meet need to be joined consistently. R packages such as SPECTROLAB (Meireles *et al.*, 2017) have options for resampling, trimming, and joining data between the silicon and InGaA sensors.

The success of aggregating herbarium reflectance spectra will depend on practices that ensure consistency and interoperability across institutions, including data quality controls and removal of low-quality spectra. Routine validation processes will ensure that data meet the necessary standards for reproducibility and analysis.

Some of these issues were encountered and overcome by a National Institute for Mathematical and Biological Synthesis working group on the evolution of leaf spectra (Meireles *et al.*, 2020a), which harmonized spectral data collected from fresh leaves using different instruments and protocols around the world. They were able to reconcile instrument-specific variation by measuring a subset of tissues with multiple instruments, using these paired measurements to correct for variation, and converting the spectral data to a common format. Their solutions demonstrated the promise of aggregating spectral measurements by showing that reflectance signatures are tightly coupled to the tree of life and have the capacity to reveal evolutionary history. To ensure long-term utility and transparency, raw spectral data should always be archived in the original format collected by the instrument. This preserves the full dataset and avoids irreversible alterations introduced by post-processing steps.

6. Fitting appropriate functional trait models to specimens

A key consideration is the transferability of spectral models between fresh, pressed, and herbarium specimens. While recent studies have confirmed that these specimen types are all informative for functional trait prediction and taxonomic classification (Wagner *et al.*, 2019; Kothari *et al.*, 2023), accounting for minor spectral shifts due to aging and preservation during model building will be important. Robust trait prediction models will need to include a wide range of species, leaf ages, drying temperatures, preservation techniques, and storage conditions. Deciphering and accounting for these influences is an active area of investigation. Spectral models trained on recently pressed specimens may perform well on herbarium specimens, but in some cases, degradation will likely bias trait modeling (e.g. Chl). There is less reason to expect models to transfer well between fresh and dried leaves since drying involves major changes in structure and water content. Developing correction factors and harmonization techniques for cross-specimen spectral applications will be necessary to expand the utility of herbarium spectral databases. When seeking to apply a trait model, some ground-truthed data will be needed to ensure accurate predictions for a particular class of specimens based on taxa sampled, the herborization process, and instrumental setup. Without validation, there is no basis for ensuring that models developed for one set of specimens will accurately predict traits in another set of specimens.

IX. Database sufficiency

The development of cyberinfrastructure has been pivotal in enabling large-scale aggregation of spectral data. Platforms such as iDigBio, *speciesLink*, and GBIF provide centralized repositories for biodiversity data, but dedicated cyberinfrastructure for spectral datasets, integrated with existing platforms, will be essential for advancing collections-based research. These systems should support real-time synchronization of available data from herbaria, cross-referencing, and retrieval for global accessibility. Any dedicated spectral cyberinfrastructure platforms will require Application Programming Interfaces to enable researchers to

query, retrieve, and contribute spectral datasets programmatically, facilitating the large-scale synthesis of data. While several cyberinfrastructure platforms have been developed specifically for spectral data and models, such as EcoSIS and EcoSML (Wagner *et al.*, 2019), they are not appropriate or sufficient for capturing the taxonomic information required in biodiversity collections. A more efficient solution will be to ensure that the databases from natural history museums can incorporate spectral data and the critical associated metadata.

On top of this, implementing analysis engines capable of handling high-dimensional datasets will be transformative. These engines should be able to integrate spectral data with complementary datasets, such as genomic or spatial data, and provide tools for advanced modeling and visualization. Open-source analysis platforms with user-friendly interfaces will democratize access to these tools and foster collaboration across disciplines.

1. Darwin Core

The Darwin Core (DC) standard is a compendium of terms and definitions that describe the observation of an organism at a particular time and space (Wieczorek *et al.*, 2012). Broadening DC to incorporate herbarium reflectance spectroscopy will enable data sharing across existing biodiversity informatics platforms and the use of spectral information in biodiversity research and conservation. In addition, extending DC for spectroscopy will supply new means for sharing protocols and practices for data aggregation to improve the quality of the herbarium reflectance data collection, storage, and distribution.

2. New specimen collections and opportunities for improved spectral digitization practices

We have thus far emphasized the role of previously collected herbarium material in generating taxonomic and trait information from the global metaherbarium. New specimen collections that fill geographic and taxonomic gaps are equally important (Ondo *et al.*, 2024). For new collections, there is the opportunity to establish best practices for spectral digitization, including measurement before mounting and minimal impact specimen preservation, to avoid many of the issues of specimen damage, paper background, glue, and degradation of material due to aging, preservation techniques, and storage conditions.

X. Effective, equitable, and ethical sampling

Herbarium users and stewards must balance innovation with preservation. Guidelines for the effectiveness and ethics of our global biodiversity heritage have been recently formalized to better utilize and steward these collections and safeguard their continued use (Davis *et al.*, 2025). Many of these guidelines have already been in place for decades and are often included in institutional policies that have been deliberated by stewards and shared with users (Rabeler *et al.*, 2019; Shah, 2023). The herbarium and natural history museum community has become increasingly aware of what we call the ‘destructive sampling conundrum’: How

to foster innovative research that includes destructive sampling of specimens that are meant to be protected permanently (Davis *et al.*, 2025). We additionally recognize that herbaria are differentially concentrated around the world, especially the global north (Heberling *et al.*, 2019; Park *et al.*, 2023), and are of different sizes and face a range of operating circumstances and challenges (National Academies Geof Sciences, Engineering and Medicine, 2020).

The effective implementation of spectroscopic sampling and measurement strategies requires meticulous consideration of both the specimen’s characteristics and the preservation and repair methods that the world’s herbaria employ. Under ideal conditions, spectral digitization provides a nondestructive alternative to traditional trait measurements, allowing researchers to infer key plant traits without physically altering specimens (Kühn *et al.*, 2024; White *et al.*, 2025b). By nondestructive, we mean the specimen is left the way it was previously with no trace of tissue removal (Davis *et al.*, 2025). Spectral data can be collected non-destructively when either detached leaves are available or when some leaves on a specimen are not fully glued. Damage or removal of specimen tissue for spectroscopy efforts is likely to occur if nondestructive protocols are not enforced. While spectral measurements can be made nondestructively, testing the accuracy of trait predictions requires some amount of trait validation, which implies the destructive use of tissue from similar taxa ideally with similar herborization processes. The high performance of trait models from pressed leaves (Kothari *et al.*, 2023) applied to herborized specimens (White *et al.*, 2025b) is a promising indicator that models validated against pressed leaves can also be used to model herbarium leaf tissues representing similar taxonomic and trait spaces. If tissue removal from specimens is required, it should be absolutely minimal, avoiding key features for identification or possible future use, and documenting what was removed. It is crucial to seek permission for these efforts. To minimize the need for destructive sampling, unwanted duplicates could be retained for spectroscopic and destructive trait measurements to calibrate trait models. Alternatively, excess duplicate specimens could be set aside and offered to the network of herbarium spectroscopy researchers, advancing the development of spectroscopic libraries for trait estimation. Adhering to nondestructive measurement approaches should be strongly encouraged to maintain the structural and historical integrity of the specimens. Researchers should obtain explicit permission before conducting studies and ensure that their methodologies align with the conservation objectives of the hosting institutions.

Finally, it is important that any databased content or spectral signatures harvested be offered as shared data to the host institution as part of the global metaherbarium, making the data available immediately. Doing so will help to avoid duplication of efforts for replicated specimens at other institutions. Finally, it is essential to promote collaboration, including by recognizing herbaria scholars, staff, and scientists formally as contributors, collaborators, funded partners, and coauthors (Edwards *et al.*, 2024; Davis *et al.*, 2025) following practices that require extraction of tissue or other alterations to herbarium specimens (e.g., Burbano & Gutaker, 2023; Davis, 2023; Davis & Knapp, 2025). Establishing

clear guidelines for sampling ensures that scientific investigations do not compromise the preservation of these collections. For example, the American Society of Plant Taxonomists has outlined best practices that emphasize the importance of minimizing physical alterations to specimens (Rabeler *et al.*, 2019). By adhering to such protocols, researchers can foster trust and facilitate ongoing access to these vital resources. All of these considerations are paramount in the application of leaf spectroscopy to herbarium specimens and for effective, equitable, and ethical sampling strategies (Davis *et al.*, 2025). Sharing findings and derivative data with the broader scientific community will promote transparency and collective advancement of the plant sciences.

XI. Next steps

In closing, we declare that the international herbarium spectral digitization (IHerbSpec) working group is ready for next-generation spectral digitization. We acknowledge that we have considerable work to do as a community to ensure that data can be aggregated and to ensure fair and equitable practices. As a first step, we have developed an initial protocol and metadata standards for the community. IHerbSpec welcomes new collaborators as we endeavor to map out the future of spectral digitization. We pledge to be collaborative, inclusive, and equitable in our work and to generate data from the world's herbaria to advance understanding and protection of the world's biodiversity.

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Author contributions

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Table S1 Previous studies using dry leaf spectra.

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